MIL-C-12386E
31 March 1987
SUPERSEDING
MIL-C-12386D
24 July 1979

MILITARY SPECIFICATION

CLAMP, REPAIR, PIPE:

WITH GASOLINE-RESISTANT GASKET,

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a single half-shell metal clamp with stirrup bolt and split- or pinhole-type flexible gasoline-resistant gasket for application over damaged pipelines.

1.2 Classification. Clamps shall be of the following types, lengths, and sizes, as specified (see 6.2):

Type I - Split application. (MIL-C-12386/1)
Lengths - 3-, 6-, 9-, and 12-inch.
Sizes - 4-, 6-, 8-, and 12-inch diameter pipe.

Type II - Pinhole application. (MIL-C-12386/2)
Length - 3-inch.
Sizes - 4-, 6-, 8-, and 12-inch diameter pipe.

1.2.1 Part number. Specification part numbers for items described in this specification will be identified as shown in the applicable specification sheets, MIL-C-12386/1 and MIL-C-12386/2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730

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Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-S-1362	- Stud, Plain; General Purpose.
FF-W-92	- Washer, Flat (Plain).
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-1086	- Gasket Material, Synthetic Rubber (for Bolted Steel Tanks).
MIL-C-12386/1	- Clamp, Repair, Pipe; with Gasoline-Resistant Gasket; Type I, Split Application.
MIL-C-12386/2	- Clamp, Repair, Pipe; with Gasoline-Resistant Gasket; Type II, Pinhole Application.
MIL-C-46168	- Coating, Aliphatic, Polyurethane Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practices for Commercial Packaging.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 Structural Welding Code - Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box, 351040, Miami, FL 33126.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for specifications sheets), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Clamps with gaskets shall be as shown on the applicable specification sheet (MIL-C-12386/1 and MIL-C-12386/2) and as specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of clamps from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish clamps conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The pipe shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated

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to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the clamps may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4 Shells and lugs. Shells and lugs shall be steel not less than 0.180 inch thick conforming to QQ-S-698, hot-rolled commercial quality sheet or strip with mill edge. Curvature of the shell shall conform to the nominal radius of the specified pipe size.

3.4.1 Length of shell. Basic shell length shall be 3 inches for all clamps. The length of type I clamps shall be increased by 3-inch increments to obtain longer lengths.

3.4.2 Lugs. Lugs shall be either welded to or formed as part of the shell body and shall develop the full strength of the clamp. Each 3-inch segment of shell shall have lugs for a stirrup bolt.

3.5 Stirrup bolts. Stirrup bolts shall be fabricated from carbon or alloy steel studs conforming to FF-S-1362. The stirrup bolts shall have equivalent coarse series thread reversed on opposite ends and shall be cut, rolled, or ground to a length allowing not less than two full turns of the nut past its fully tightened position for the pipe size of the clamp. Curvature of the formed stirrup bolt shall be the same as curvature of the shell.

3.5.1 Sizes of bolts. Bolt sizes shall conform to table I. The inner surface of the stirrup bolt shall be flattened to provide a bearing surface on the pipe. The flattened surface shall be not less than 0.5 inch wide and shall provide an equal and even bearing on the pipe. Each type II clamp shall be furnished with one bolt. The number of bolts furnished with each type I clamp shall be as follows:

Clamp length	Number of bolts
3 inches	1
6 inches	2
9 inches	3
12 inches	4

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TABLE I. Pipe and stirrup bolt sizes.

Clamp (nominal pipe size)	Outside diameter of pipe	Stirrup bolt sizes
Inches	Inches	Inches
4	4.500	1/2 by 12
6	6.625	5/8 by 18
8	8.625	5/8 by 24
12	12.750	5/8 by 36

3.6 Nuts. Nuts for stirrup bolts shall conform to FF-N-836, type II, style 4, grade A steel, coarse thread series and sizes to match the bolt sizes specified in table I.

3.7 Washers. Washers shall be furnished with all stirrup bolts and shall conform to FF-W-92, type A, grade I, class A, sizes to match the bolt sizes specified in table I.

3.8 Hydrostatic pressure. Clamps shall not leak when subjected to the hydrostatic pressure specified for each size clamp, as follows:

Clamp (nominal pipe size)	Pressure pounds per square inch (psi)
4 inches	800
6 inches	740
8 inches	570
12 inches	510

3.9 Identification marking. Clamps shall be identified by rolling, stamping, or stenciling in accordance with MIL-STD-130.

3.10 Treatment and painting. Unless otherwise specified (see 6.2), all surfaces of the clamps, normally painted internally and externally shall be cleaned, treated and painted in accordance with MIL-T-704, type F and top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.11 Type I clamps. Clamps for application over splits shall be furnished with sheet gaskets conforming to MIL-G-1086, type I. Gaskets shall be 0.125 inch thick and shall be not less than half the diameter of the pipe size in width. The length of the gasket shall be the same as the length of the shell.

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3.12 Type II clamps. Clamps for application over pinholes shall be furnished with molded gaskets conforming to MIL-G-1086, type II. Gaskets shall be square or cone shaped. Square gaskets shall be 0.125-inch thick by 1.625-inch square. Cone shaped gaskets shall be 1.625-inch in diameter by 0.125-inch thick at the edges and 0.375-inch thick at the point of the cone.

3.13 Workmanship.

3.13.1 Steel fabrication. Metal used in fabrication shall be free from kinks and sharp bends. Straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the metal, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated metal shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.13.2 Boltholes. Boltholes shall be accurately formed and shall have the burrs removed.

3.13.3 Welding. Surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Weldments including weld beads, shall be of homogeneous appearance and form without spattering or other irregularities. The weldment shall comply with the size, configuration and other dimensional requirements to develop the full strength of the parts joined by the welds. No cracks of any type are permitted.

3.13.4 Welders. Before assigning any welder or welding operator for welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code - Steel, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". The certification shall be made available for review by the contracting officer or the contracting officer's representative. The Government reserves the right to require recertification of any welder or welding operator.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all

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inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article clamp inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article clamp inspection.

4.3.1 Examination. Clamps shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Clamps shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot. For the purpose of inspection, a lot shall consist of all types, lengths, and pipe sizes of clamps offered for delivery at one time.

4.4.2 Sampling. Sampling for examination shall be in accordance wilth MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined as specified in 4.5.1. Acceptable Quality Level (AQL) shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.4 Tests. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2. Inspection level shall be S-2. AQL shall be 1.0 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. Clamps shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the repair clamps.
105. Dimensions or components not in accordance with applicable specification sheet.
106. Components missing. Each missing component shall constitute one failure.
107. Clamps not of types, lengths, and sizes specified.
108. Workmanship not as specified.
109. Treatment and painting not as specified.
110. Paint color not as specified.
111. Welding not as specified.
112. Contrator does not have documentation available for materials, materials finishes or treatments.

Minor

201. Identification marking missing, incomplete, or illegible.

4.5.2 Tests.

4.5.2.1 Hydrostatic tests. Clamps shall be subjected to the following hydrostatic tests as applicable. The test fluid shall be water.

4.5.2.1.1 Type I (split application). Test as follows: (a) In a pipe of the correct size for the clamp, prepare a slit 2 inches long and 0.063 inch wide. (b) Apply clamp with gasket over the slit. (c) Raise the water pressure to the applicable hydrostatic test pressure as specified in 3.8 and hold for 30 seconds. Evidence of leakage shall constitute failure of this test.

4.5.2.1.2 Type II (pinhole application). Test as follows: (a) In proper size pipe, prepare an artificial leak 0.25 inch in diameter. (b) Apply clamp with gasket over the leaking area. (c) Raise the water pressure to the applicable hydrostatic pressure specified in 3.8 and hold for 30 seconds. Evidence of leakage shall constitute failure of this test.

4.5.2.2 Effect of hydrostatic pressure on welds. Following hydrostatic test, examine all clamp welds. Evidence of deformation or fracture of the welds shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

113.	Gaskets, nuts and washer not preserved as specified for level A.
114.	Consolidation not as specified for level A.
115.	Preservation and packing not in accordance with the referenced document specified for commercial.
116.	Quantities packed together exceed the weight or size limitations of the box for level A or B.
117.	Boxes not as specified for level A or B.
118.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Gaskets. Each gasket shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.2 Nuts and washers. Required nuts and washers for each clamp shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3; or assembled on the bolt.

5.1.1.3 Consolidation. Components comprising a complete clamp shall be consolidated together in a close fitting box conforming to PPP-B-636, W6c, style optional. Cushioning shall be provided to prevent movement within the box. Box closure and sealing shall be in accordance with method V, of the appendix to the box specification.

5.1.2 Commercial. Each clamp shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

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5.2.1 Level A. Clamps of like size and decription, preserved as specified in 5.1, shall be packed together in close fitting boxes conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitations of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Clamps of like size and description, preserved as specified in 5.1, shall be packed together as specified in 5.2.1, except that the boxes shall be domestic type or class as applicable. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c, may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.3 Commercial. Clamps preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or level B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, each shipping container shall display weight and cube data.

6. NOTES

6.1 Intended use. Clamps are intended to be used for emergency repair of pinholes and splits in pipelines.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Definitive part number of clamp, including type, size, and length required (see 1.2 and applicable specification sheet).
c. Time frame required for submission of first article model, and number of clamps required (see 3.2).
d. Treatment, painting, or color if other than as specified (see 3.10).
e. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3. First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

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6.4 Subject term (key word) listing.

Clamp, pipe
Clamp, repair, pipe
Repair clamp, pipe

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME
Air Force - 99

Review activity:
Air Force - 82
DLA - CS

User activities:
Navy - MC

Preparing activity:
Army - ME

Project 4730-0726

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-12386E	Clamp, Repair, Pipe: With Gasoline-Resistant Gasket, General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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